Capstone Problem Deliverables

A full submission will be a written report. The document should be written as an engineer making a report and recommendation to a client.

The body of the report must be suitable for reproduction, that is, computer-generated and in PDF format.

The report must contain the following deliverables, organized and ordered as follows:

1. Letter of Transmittal (on a separate page).
2. Cover Page, including the title “Technical and Economic Proposal” and the specific title of your project.
3. Executive Summary (500-600 words, on a separate page).
4. Table of Contents (on a separate page).
5. Table of Figures (on a separate page).
6. Table of Tables (on a separate page).
7. Brief Design Description (500-600 words, (on a separate page).
8. Process Details.
    a. Process flow diagram (PFD, CHEMCAD) for process and utilities.
    b. Sized equipment list (as in DP5).
9. Economics.
    a. Socio-economic analysis
    b. Capital cost estimate and table of equipment (as in DP5).
    c. Payback period, percent return on investment.
    d. Net present worth and internal rate of return.
    e. Comparison of PBP, ROI, NPW and IRR to appropriate benchmarks.
10. Process Safety.
    a. Safety and environmental impact assessment.
    b. Management of change proposal.
    c. CHEMCAD environmental report.
    d. P&ID with controls and alarms.
    e. Pressure relief valve sizing.
11. Conclusions.
12. Appendices.
13. Additional descriptions of these sections and how to construct them can be found in Chapter 11 of Peters, Timmerhaus, and West.
14. Final submission must be in electronic format (PDF).